

## Descriptives: combined\_\_9\_\_17\_\_26

The full medical-device 10-K/10-Q panel (one row per filing) with FDA product events and event 8-Ks merged in by gvkey × reporting year × reporting quarter.

Source: `data/combined_9_17_26.csv`.

Built by `scripts/18_build_combined_filing_event_dataset.py`.

### Overview

| Statistic                                                | Value                    |
|----------------------------------------------------------|--------------------------|
| Filings = firm reporting quarters (rows)                 | 6,547                    |
| 10-K                                                     | 1,660                    |
| 10-Q                                                     | 4,887                    |
| Firms (gvkey)                                            | 117                      |
| Reporting periods ending                                 | 2004-11-27 to 2026-05-31 |
| Filings with a parsed MD&A section                       | 6,238                    |
| Filings with a warning letter in the quarter             | 92                       |
| Filings with a recall in the quarter                     | 1,376                    |
| Filings with an adverse-event report in the quarter      | 4,059                    |
| Filings with no product event of any type in the quarter | 2,367                    |
| Filings with an event 8-K in the quarter                 | 74                       |

### Unique filings with a SUBSTANTIVE mention of each event in the MD&A

| Event substantively discussed in MD&A | 10-K filings | 10-Q filings | All filings | Share of all filings |
|---------------------------------------|--------------|--------------|-------------|----------------------|
| Warning letter                        | 135          | 461          | 596         | 9.1%                 |
| Recall (Part 806)                     | 272          | 581          | 853         | 13.0%                |
| Adverse event (Part 803)              | 108          | 166          | 274         | 4.2%                 |
| Any of the three (each filing once)   | 367          | 891          | 1,258       | 19.2%                |

*Counts are unique filings (accession numbers) across the FULL baseline panel, MD&A section only (Item 7 in a 10-K, Item 2 in a 10-Q). 'Substantive' = at least one MD&A occurrence tied to an actual event (past-tense receipt/action verb, definite reference, or specific date) and not hedged by a modal - script 17's definition. A filing can count under more than one event type.*

### Substantive MD&A disclosure: quarters WITH vs WITHOUT the event

| Event type               | Filings: event in quarter | ... substantive MD&A | Rate  | Filings: no event in quarter | ... substantive MD&A | Rate  |
|--------------------------|---------------------------|----------------------|-------|------------------------------|----------------------|-------|
| Warning letter *         | 91                        | 34                   | 37.4% | 5,117                        | 473                  | 9.2%  |
| Recall (Part 806)        | 1,376                     | 291                  | 21.1% | 5,171                        | 562                  | 10.9% |
| Adverse event (Part 803) | 4,059                     | 199                  | 4.9%  | 2,488                        | 75                   | 3.0%  |

*Raw rates, no controls - descriptive only. 'Event in quarter' = at least one event of that type dated inside the filing's own reporting period. Discussion in a no-event quarter is mostly the continuing discussion of an EARLIER quarter's event, so the right-hand rate is not a clean counterfactual. \* Warning-letter row uses only filings whose whole period falls on/after 2008-10-01, when FDA's letter data begins; earlier filings have letters UNOBSERVED, not absent, and are excluded rather than counted as no-event.*

### Substantive MD&A disclosure at the YEARLY (10-K) level: fiscal years WITH vs WITHOUT the event

| Event type               | 10-Ks: event in fiscal year | ... substantive MD&A | Rate  | 10-Ks: no event in fiscal year | ... substantive MD&A | Rate  |
|--------------------------|-----------------------------|----------------------|-------|--------------------------------|----------------------|-------|
| Warning letter *         | 86                          | 21                   | 24.4% | 1,184                          | 93                   | 7.9%  |
| Recall (Part 806)        | 626                         | 144                  | 23.0% | 1,034                          | 128                  | 12.4% |
| Adverse event (Part 803) | 1,166                       | 90                   | 7.7%  | 494                            | 18                   | 3.6%  |

10-K filings only. 'Event in fiscal year' = at least one event of that type dated anywhere inside the fiscal year the 10-K covers (previous 10-K's period end to this one's; all four quarters, not just Q4). Raw rates, no controls. \* Warning-letter row uses only filings whose whole period falls on/after 2008-10-01, when FDA's letter data begins; earlier filings have letters UNOBSERVED, not absent, and are excluded rather than counted as no-event.

## Event 8-Ks by event type

| 8-K event type            | Documents<br>(forms +<br>exhibits) | Unique 8-Ks | ... at<br>baseline-<br>panel firms | ... merged<br>to a<br>reporting<br>quarter | Filings<br>(quarters)<br>with such an<br>8-K |
|---------------------------|------------------------------------|-------------|------------------------------------|--------------------------------------------|----------------------------------------------|
| Warning letter            | 151                                | 115         | 65                                 | 62                                         | 60                                           |
| Recall (Part 806)         | 72                                 | 62          | 16                                 | 15                                         | 15                                           |
| Adverse event (Part 803)  | 7                                  | 6           | 5                                  | 5                                          | 5                                            |
| Any event (each 8-K once) | 222                                | 176         | 80                                 | 76                                         | 74                                           |

Source: the device 8-K dataset (script 16), classified on the filing lead. One 8-K can carry several documents (form + exhibits) and more than one event flag. Only 8-Ks filed by firms in the 10-K/10-Q baseline panel can merge; an 8-K is placed by its reported event date.

## Product event dates (the events themselves - no filings involved)

| Event type - scope                                       | Events     | Distinct<br>firm-dates | Firms | First event<br>date | Last event<br>date |
|----------------------------------------------------------|------------|------------------------|-------|---------------------|--------------------|
| Warning letter - all gvkey-linked firms                  | 192        | 188                    | 135   | 2008-10-10          | 2026-04-30         |
| Warning letter - firms in the 10-K/10-Q panel            | 128        | 125                    | 86    | 2008-10-10          | 2026-04-30         |
| Warning letter - merged to a reporting quarter           | 96         | 93                     | 67    | 2008-10-21          | 2026-01-29         |
| Recall (Part 806) - all gvkey-linked firms               | 17,448     | 5,411                  | 113   | 2000-04-11          | 2026-06-30         |
| Recall (Part 806) - firms in the 10-K/10-Q panel         | 11,722     | 3,989                  | 82    | 2000-04-11          | 2026-06-30         |
| Recall (Part 806) - merged to a reporting quarter        | 7,761      | 2,544                  | 74    | 2004-09-20          | 2026-03-30         |
| Adverse event (Part 803) - all gvkey-linked firms        | 10,737,528 | 266,574                | 177   | 1991-12-31          | 2026-06-30         |
| Adverse event (Part 803) - firms in the 10-K/10-Q panel  | 9,588,496  | 202,984                | 113   | 1991-12-31          | 2026-06-30         |
| Adverse event (Part 803) - merged to a reporting quarter | 7,114,479  | 118,510                | 110   | 2004-09-03          | 2026-05-28         |

Counted from the FDA source data. Event = one warning letter (unique Case ID, dated by action date), one recall record (dated when the firm initiated it), one adverse-event report (dated when FDA received it). 'Distinct firm-dates' counts each firm x calendar date once. Events not merged belong to firms outside the 10-K/10-Q panel, predate the firm's first filing in it (the panel starts in 2005; MAUDE in 1991), postdate its last, or fall in a gap between filings.

## Industry coverage: panel share of the US publicly traded medical-device industry

| Statistic                                                   | Value                    |
|-------------------------------------------------------------|--------------------------|
| Firms in the 10-K/10-Q panel                                | 117                      |
| in the US medical-device universe (counted in the share)    | 31                       |
| no current market cap (acquired, delisted or private since) | 53                       |
| current market cap, but primary SIC outside 3841-3845       | 30                       |
| device SIC, but incorporated outside the US                 | 3                        |
| US medical-device universe, firms                           | 167                      |
| Panel firms as a share of universe firms                    | 18.6%                    |
| Share of US medical-device market capitalization            | 81.0%                    |
| Market cap of panel firms in the universe (\$M), mean       | 29,794                   |
| Market cap of panel firms in the universe (\$M), median     | 3,161                    |
| Total assets of panel firms in the universe (\$M), mean     | 10,397                   |
| Total assets of panel firms in the universe (\$M), median   | 1,827                    |
| Fiscal year-ends at which market cap is measured            | 2025-06-30 to 2026-06-30 |

*Share = current market capitalization of the panel firms that are in the US medical-device universe, divided by the universe total. Universe: US-incorporated Compustat firms with primary SIC 3841-3845 and a computable market cap (fiscal year-end price  $\times$  shares outstanding) at their latest fiscal year-end on or after 2025-06-30 - the definition used for Table 1 of 2026-07-06, recomputed here for the 10-K/10-Q panel. It is a CURRENT snapshot: panel firms acquired or delisted since they entered the sample contribute nothing, and neither do panel firms classified outside the device SIC codes or incorporated abroad, however large - so the share understates the panel's historical coverage. Source: Compustat (comp.company, comp.funda) via WRDS.*

## Caveats

1. Reporting year/quarter is the calendar year and quarter of the period end (snapped to the nearest month end) - a label for the firm's reporting period, not its own fiscal-quarter number. A 10-K row is the fiscal 4th quarter.
2. An event is attached ONLY to the quarter it is dated in. Firms keep discussing a warning letter for years, so lagged/cumulative event indicators are needed before reading the no-event rate as a counterfactual.
3. Adverse-event reports arrive almost every quarter at large firms, so event\_adverse\_event has little variation there; use the counts (n\_adverse\_event, and n\_mdr\_tier\_a / n\_mdr\_tier\_b separately - Tier B is parent-prefix name matching).
4. Events use script 14's ownership-window gate but NOT its comp.funda gate; compustat\_active\_year flags the firm-years that gate would keep.
5. Only Risk Factors and MD&A are parsed (script 17); event discussion located only in Legal Proceedings (Item 3) is missed.